

## Sensitivity and Stability Analysis

This document provides additional analyses on the sensitivity and stability of the proposed algorithms. We first examine the hyperparameter sensitivity of the EM algorithm with respect to the number of prompts ( $K$ ) on the World Values Survey dataset (Table D.13). We then evaluate the sensitivity to meta-prompt wording by paraphrasing the original meta-prompt templates using Gemini 3.1 Pro and rerunning the experiments with all other procedures fixed (Table D.14). Finally, we assess algorithmic stability across multiple independent runs, where both EM and GB are evaluated on the MobLab dataset using five random seeds (Table D.15).

Table D.13. Hyperparameter sensitivity analysis of the GB algorithm on the WVS dataset with different numbers of prompts ( $K$ ).

| Configuration | W-dist | Mean  | Std   |
|---------------|--------|-------|-------|
| $K = 10$      | 1.92   | 42.58 | 17.63 |
| $K = 40$      | 1.58   | 44.60 | 19.09 |

Table D.14. Meta-prompt sensitivity analysis on the WVS dataset.

| Meta-prompt Template | EM   | GB   |
|----------------------|------|------|
| Original template    | 2.15 | 3.34 |
| Paraphrased template | 2.72 | 2.19 |

Table D.15. Statistical reporting over five independent runs on the MobLab dataset. We report the mean and standard deviation of the Wasserstein distance.

| Method | Dictator        | Proposer        | Responder       | Investor        | Banker          | Public Goods    | Bomb            |
|--------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| EM     | $1.86 \pm 0.59$ | $2.79 \pm 0.86$ | $3.39 \pm 1.12$ | $2.33 \pm 0.56$ | $6.82 \pm 2.29$ | $0.72 \pm 0.14$ | $7.78 \pm 2.66$ |
| GB     | $1.95 \pm 0.12$ | $1.88 \pm 0.18$ | $2.64 \pm 0.08$ | $2.26 \pm 0.39$ | $4.81 \pm 0.97$ | $1.00 \pm 0.25$ | $6.18 \pm 1.28$ |